

Argument Graph

A tool for dissecting the logical structure and evidential support of research papers
Product Requirements Document • Draft v0.1

1. Summary

Reading a paper critically means reconstructing an argument the paper never states explicitly: which claim is the thesis, which claims it rests on, what evidence is offered for each, and where the inferential joints are weakest. Experienced reviewers do this in their heads and lose the work the moment they close the PDF. Everyone else does it partially, or not at all.

Argument Graph parses a paper into an explicit graph of claims, evidence, and support relations, then renders that graph in a GUI built for interrogation rather than reading. The unit of analysis is the **claim**; the unit of output is a **traceable assessment of how well each claim is supported**, decomposed into named dimensions rather than collapsed into a single score.

The product is a reasoning aid, not a verdict machine. Every assessment is anchored to a text span in the source and is editable by the user. A rendered graph the reader disagrees with is a success case: it makes the disagreement locatable.

2. Users and jobs

User	Job to be done	Success looks like
Peer reviewer	Assess whether conclusions are licensed by the evidence, under time pressure.	Weakest inferential links surfaced in minutes, with spans to cite in the review.
Grad student / early researcher	Read critically in an unfamiliar subfield without knowing the conventions yet.	Sees the argument skeleton and where the field's standard moves are being made.
Systematic reviewer / meta-analyst	Screen many papers for claim-evidence fit and scope creep.	Comparable structured extractions across a corpus, exportable.
Journalist / policy analyst	Determine what a paper actually establishes vs. what its abstract implies.	Explicit gap between headline claim scope and supported claim scope.

3. Core model: the argument graph

Everything downstream is a projection over one immutable artifact. Extraction produces a graph; the GUI, the scores, and the exports are all read-only functions of it. Corrections are appended as events rather than mutating the extraction, so the model's output and the user's judgment stay separable and the graph at any point is a fold over that event log.

```
graph = fold(apply_edit, extract(paper), edits)
```

3.1 Node types

Node	Definition	Key attributes
Claim	An assertion the paper makes or relies on.	kind (empirical / causal / definitional / normative / methodological); scope; hedging level; is_thesis; span
Evidence	A result, dataset, figure, table, or statistical test offered in support.	type; effect size; uncertainty; n; span
Assumption	A proposition required for an inference to hold, often unstated.	explicit vs. inferred; criticality; span (nullable)
Citation	An external work invoked as support.	DOI; claimed content; verification status
Method	A procedure whose validity conditions gate the evidence.	design; preregistered; controls; span

3.2 Edge types

- **supports** — source provides evidential warrant for target. Weighted; the weight is the product of the support dimensions in §4.
- **undermines** — source counts against target (acknowledged limitations, contrary results, failed robustness checks).
- **depends_on** — target is not licensed unless source holds. Assumptions attach here; this is the edge that carries fragility.
- **elaborates** — target restates or narrows source. Used to detect scope drift between the abstract and the results.
- **cites** — claim invokes an external work. Verification is a separate pass (§6.3).

The graph is a DAG over claims with cycles permitted only through undermines. Topological order gives the natural reading order for the argument, which is usually not the paper's section order — that difference is itself informative and is surfaced in the GUI.

4. Support strength: a decomposition, not a score

A single 0–100 rigor number would be the fastest way to make this tool both wrong and popular. Instead each supports edge carries independent dimensions, each with its own confidence, each traceable to a span. The GUI shows the vector; aggregate rollups are available but always expandable to the components.

Dimension	Question it answers	Failure it catches
Directness	Does this evidence bear on this claim, or on a neighboring one?	Evidence for a proxy outcome presented as evidence for the outcome of interest.
Inferential gap	How far must you travel from the evidence to the claim?	Correlation stated as causation; mechanism inferred from association.
Scope match	Does the claim's generality match the sample or setting?	Population claims from a convenience sample; species / context over-extension.
Statistical support	Do the reported statistics license the strength asserted?	Underpowered designs; p-values without effect sizes; uncorrected multiplicity.

Dimension	Question it answers	Failure it catches
Assumption load	How many unstated propositions must hold?	Long dependency chains where one weak link collapses the conclusion.
Citation integrity	Does the cited source say what it is claimed to say?	Citation drift; propagated claims that were hedged in the original.
Hedging fidelity	Does the language match the evidential strength?	Hedged in the results, asserted flatly in the abstract and title.

Fragility is the derived, headline metric: for each claim, the minimum-weight cut through its support subgraph. It answers “what is the smallest thing that, if wrong, breaks this conclusion?” and it points at a specific node rather than producing an adjective.

5. Interface

Four coordinated views over the same graph, with selection state shared across all of them. Selecting a claim anywhere highlights its span in the text, its node in the graph, and its row in the weak-links list.

5.1 Argument map (primary)

- Thesis at the root, support flowing upward; layout is a function of the graph, not hand-positioned.
- Edge thickness encodes support weight; edge color encodes supports vs. undermines vs. depends_on.
- Nodes collapse to subtrees so a 200-node paper stays legible; default view shows depth 2 from thesis.
- Assumption nodes render distinctly and are togglable — hiding them is the “as the authors present it” view, showing them is the “as it must actually hold” view.

5.2 Reading view

- The paper text with claim spans underlined and gutter markers for support strength.
- Hovering a claim dims everything except its evidence spans — including evidence in distant sections, which is where claim–evidence distance becomes visible.
- Abstract-vs-body diffing: claims in the abstract are matched to their body counterparts and scope drift is flagged inline.

5.3 Claim inspector

- Full dimension breakdown for the selected claim with the span backing each judgment.
- Every dimension is user-overridable, with a note field; overrides propagate to fragility immediately.
- “What would change this?” panel: the specific evidence that would raise the weakest dimension.

5.4 Weak links

- A ranked list of the paper's load-bearing weak points, sorted by fragility contribution.
- This is the review-writing surface: select rows, export as structured comments with spans and page refs.

6. Pipeline

pdf → layout → segments → candidate claims → relations → dimensions → graph → views

6.1 Ingest and segmentation

PDF layout extraction with section, figure, table, and reference-list identification. Segment to sentence level with stable character offsets — every downstream artifact references offsets, so provenance survives re-extraction. Papers with structured full text available (JATS, LaTeX source) skip the lossy layout step and should be preferred as inputs.

6.2 Extraction

LLM extraction of claims, relations, and assumptions under a constrained schema, with two hard requirements: every node carries the span it came from, and the model may not assert a relation between nodes it did not extract. Extraction runs section-scoped for recall, then a reconciliation pass merges duplicate claims across sections. Self-consistency across multiple samples produces the extraction confidence shown in the GUI.

6.3 Citation verification (phase 2)

The citation integrity dimension requires reading the cited work, which makes it the only part of the system gated on corpus access. Scoped to legally available sources: open-access full text via Unpaywall / OpenAlex / arXiv / PMC, abstracts elsewhere, and “unverifiable — no accessible full text” as an explicit, visible state rather than a silent pass. Roughly half of recent literature resolves to a legal full text; the honest UI move is to show coverage rather than imply completeness.

6.4 Persistence

Graph plus append-only edit log per paper. Extraction is versioned by model and prompt revision so a re-run is a new derivation rather than an overwrite, and user edits rebase onto it. Export to JSON, and to a flat claim–evidence table for corpus-level analysis.

7. Non-goals

- **Not a fraud or misconduct detector.** Data fabrication and image manipulation are a different problem with different methods.
- **Not a replacement for domain expertise.** The tool assesses argument structure, not whether a field's standard assumptions are sound.
- **Not a paper score or ranking.** No leaderboard, no letter grade, no single number exported for someone else to misuse.
- **Not a replication predictor.** Tempting adjacency, entirely different evidence base.
- **Not a literature search tool.** Assumes the user already has the paper.

8. Risks

Risk	Why it is serious	Mitigation
Plausible-but-wrong extraction	An LLM will always produce a clean-looking argument graph, including for papers whose argument it misread. Confident structure is more dangerous than no structure.	Span grounding on every node; extraction confidence surfaced, not hidden; low-confidence regions rendered as unresolved rather than smoothed over; user override on every judgment.

Risk	Why it is serious	Mitigation
Authority laundering	Users treat a rendered graph as an objective finding and cite it as such.	No composite score. Assessment language is dimensional and hedged. Provenance is always one click away and always shown on export.
Discipline mismatch	Rigor conventions differ sharply — what counts as adequate support in ethnography, RCT medicine, and pure mathematics has almost nothing in common.	Ship one discipline family first (empirical quantitative). Discipline-specific dimension weights and rubrics thereafter. Refuse to render, rather than guess, outside supported families.
Adversarial use	A dimensional critique generator is an efficient tool for discrediting inconvenient findings.	Symmetric framing (surfaces strengths as well as weak links); export always carries the unverified / low-confidence flags.
Citation verification coverage	Partial coverage silently reads as “no problems found.”	Coverage percentage is a first-class element of the UI; unverifiable is a distinct state from verified.

9. Success metrics

- **Extraction agreement:** ≥ 0.7 Cohen's κ against expert-annotated claim/relation sets on a held-out set of 50 papers.
- **Weak-link precision:** of the top-5 fragility items per paper, the fraction domain experts agree are genuine weaknesses. Target $\geq 60\%$ at launch; this is the metric that decides whether the product is real.
- **Edit rate:** user overrides per paper. Falling over time is good; near zero is suspicious (indicates passive acceptance, which the design is meant to prevent).
- **Time to first useful insight:** under 5 minutes from upload for a 30-page paper.

10. Phasing

Phase	Scope	Exit criteria
P0 — Extraction spike	No GUI. CLI extraction to JSON on 20 papers in one subfield. Manual expert annotation for comparison.	Extraction agreement clears the κ bar. If it does not, the product does not exist yet.
P1 — Read-only viewer	Argument map + reading view with span linking. Support dimensions computed, no editing.	Five target users report the map matches their own reading of a paper they know well.
P2 — Interrogation	Claim inspector, overrides, edit log, weak-links export.	Reviewers complete a real review using the tool and keep the exported comments.
P3 — Citation integrity	OA full-text resolution and citation verification pass, with coverage reporting.	Citation drift detected at usable precision on a seeded benchmark.
P4 — Corpus mode	Batch extraction, cross-paper claim comparison, structured export for meta-analysis.	A systematic reviewer screens a real corpus faster than their current process.

11. Open questions

- What is the right granularity for a claim? Sentence-level over-fragments; paragraph-level hides the inferential joins. Likely needs to be resolved empirically in P0.
- Should implicit assumptions be extracted at all in v1? They are the most valuable output and the least reliable one.
- How are review-response papers and multi-study papers handled — one graph or several linked graphs?
- Does the graph belong to the reader or to the paper? Shared/collaborative annotation changes the data model substantially and should be decided before P2, not after.
- Which discipline family ships first, and who is the expert annotator pool for its benchmark?